

NADA MOHAMED

AI/ML Engineer

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Summary

Computer Science graduate (2026) specializing in Artificial Intelligence and Machine Learning, with an engineering background spanning intelligent agents, LLM fine-tuning, computer vision, full-stack development, and blockchain. Shipped 15+ projects end to end and taught programming to 30+ students — equally at home building systems and explaining them.

Education

Arab Academy for Science, Technology & Maritime Transport

Sep 2022 – Jun 2026

B.Sc. Computer Science (Dual Degree), Graduated with Honours – GPA: 3.69 / 4.00

Cairo, Egypt

- Dual-degree program with the University of Northampton (UK). Coursework: Machine Learning, Computer Vision, NLP, Algorithms, Parallel Computing, Compilers, Databases.

Skills

Languages: Python, Go, Java, C/C++, JavaScript, TypeScript, PHP, Dart, R, Bash, SQL, HTML/CSS

AI / ML & Data: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Librosa, HuggingFace, Whisper; NLP, Generative AI & LLM fine-tuning, Computer Vision, Reinforcement Learning, Explainable AI (SHAP, Grad-CAM)

Web & Application: Flask, FastAPI, React, Tailwind CSS, Vite, JavaFX, Tkinter, Flutter, SwiftUI, PyWebView

Systems, Cloud & DevOps: Docker, Kubernetes, Git, GitHub Actions (CI/CD), MySQL, AWS RDS, SQLite; parallel computing, compilers, algorithm design, Arduino

Blockchain & Security: Hyperledger Fabric, Chaincode (Go), X.509 PKI, cryptography, threshold (MPC) custody

Soft Skills: Mentorship & instruction, team collaboration, technical writing & documentation, analytical thinking

Experience

Dell Technologies

Aug 2025 – Sep 2025

Summer Intern

Cairo, Egypt

- Led the Academy Project end to end: scoped the idea, built the case, presented the proposal.
- Worked across sales, pre-sales, and engineering; applied design thinking to Dell's onboarding.

DEPI – Microsoft ML Engineering Curriculum

Nov 2024 – May 2025

AI/ML Engineer Intern

Zamalek, Egypt

- Completed a 7-month intensive AI/ML program covering NLP, generative AI, computer vision, and deployment.
- Delivered milestone projects and a capstone integrating multimodal workflows and agent orchestration.

Megz — IGCSE Computer Science

Aug 2020 – Nov 2021

Computer Science Instructor

Cairo, Egypt

- Taught Python and Java fundamentals to 30+ students across multiple IG schools.
- Supported the lead teacher in classroom management, grading, and student progress tracking.
- Mentored students in debugging and problem-solving; personalized guidance improved small-group performance 20%.

Projects

Payramid — Blockchain Payroll for Egypt 🌐🔒

Sep 2025 – Present

Go, TypeScript, Blockchain, Docker · Graduation Project

- Payroll orchestration and immutable audit layer letting employers disburse salaries over Egypt's national payment rails with verifiable proofs of payment, within local banking and data-protection regulation.
- Authored the thesis and the IEEE 830 SRS start to finish, from requirements through verification; currently revising the accompanying research paper.

Music Genre Detection 🎧

Dec 2025

Python, PyTorch, Librosa, Flask, Captum

- CNN-LSTM attention model classifying audio into 10 genres from mel-spectrograms, served through Flask.
- Grad-CAM and Integrated Gradients explainability, a pytest suite with coverage, and Sphinx API docs.

Pixel Paradise 🎮

May 2024 – Feb 2025

JavaScript, PHP, HTML/CSS, Python

- Retro browser arcade built with a small team over 126 commits — playable games, accounts, and score tracking behind a handcrafted pixel-art interface. Core developer across gameplay and site features.

Certifications, Awards & Languages

🏆 **First Place, AI/ML Project Submission** – AASTMT • **Microsoft ML Engineer Course** – DEPI

🏆 **AI & Machine Learning Course** – Universitat Autònoma de Barcelona • **CS Camp** – Carnegie Mellon University

Languages: Arabic (Native) • English (Fluent) • Spanish (Intermediate)